

VISHWANATH KARUNANITHI

Electronics & Communication Engineer | Embedded Systems | VLSI | IoT | Industrial Automation
visisiva.09@gmail.com | +919345247382 | Chennai, TamilNadu | linkedin.com/in/vishwanath-karunanithi-087b62308 |
github.com/vishwanathkarunanithi

PROFESSIONAL SUMMARY

Electronics and Communication Engineering student (CGPA: 8.35) with hands-on experience in Full Stack Development Embedded Systems (STM32/ESP32), VLSI Digital Design (SystemVerilog/Xilinx Vivado), and Industrial Automation (LabVIEW/myRIO). Proven track record of delivering award-winning, industry-recognized projects and securing competitive internships through hackathon victories and IIT Research Park presentations. Experienced in rapid prototyping, AI-assisted development, and real-time industrial problem-solving.

TECHNICAL SKILLS

Languages: Java, C, Embedded C, SystemVerilog, Python (Intermediate);

Embedded Platforms: STM32 (Blackpill/Bluepill), Arduino Nano, ESP32, NI myRIO

Tools & Software: Xilinx Vivado, Synopsys Tools, NI LabVIEW (CLAD certified), MATLAB, Arduino IDE, MIT App Inventor, Sinric Pro, Hugging Face

Core Domains: Embedded Systems & IoT, VLSI & Digital Design, Industrial Automation, Digital Twin Technology, AI-Assisted Development (Prompt Engineering)

Soft Skills: Rapid Prototyping, Team Collaboration, Technical Presentation, Problem Solving, Adaptability

PROFESSIONAL EXPERIENCE

Edge AI Intern (Paid — ₹10,000 Stipend) | Grupo Antolin | Feb 2026 – June 2026

Secured via IIT Research Park project recognition.

- Developing an Edge AI SOP Validator for manufacturing environments to automate quality assurance and compliance validation.
- Collaborating with industry engineers to address real-time production challenges and support AI-based workflow optimization.
- Contributing to deployment planning, industrial testing, and rapid prototyping of AI-driven industrial solutions.
- Documenting project workflows, implementation procedures, and process improvement initiatives.

Full-Stack Intern (MERN) | IBM – Naan Mudhalvan Initiative | Completed

- Developed a full-featured CRM application using the MERN stack (MongoDB, Express, React, Node.js).
- Implemented frontend UI, RESTful backend APIs, and database integration; participated in debugging and deployment activities.

Embedded Systems Intern | One Yes Info Tech | Completed

Secured as prize for winning VIT 24-Hour Hackathon.

- Developed EEPROM-based embedded system simulations and data persistence mechanisms for microcontroller applications.
- Performed debugging, validation of application logic, and documented implementation processes.

LabVIEW Intern | Aadhya Skills | 10-Day Offline Internship

- Executed hands-on experiments using NI LabVIEW and myRIO hardware platforms for industrial and medical application prototypes.
- Performed data acquisition, control-based automation exercises, and contributed to project demonstrations.
- Completed 15 real time labview projects.

KEY PROJECTS

Piracy Prevention & Detection System — *IR-LED Hardware + Software Steganography*

- Designed a dual-layer digital piracy prevention solution integrating IR-LED hardware with steganography-based software protection.
- Won ₹4,000 at TECHSTROM; invited to present at IIT Research Park, leading directly to industry internship placement.

Bio-Guardian – Smart Non-Invasive Health Monitoring — *Embedded IoT*

- Built a continuous non-invasive monitoring system for vital health parameters; showcased publicly for 3 days at Vadiva Tech Fest.
- Won First Prize (₹2,500) at Veltech Multi Tech engineering symposium.

Smart Anti-Theft Ration Distribution System (Smart-PDS) — *STM32 Microcontroller*

- Developed STM32-based automation to secure public distribution systems, preventing resource diversion and improving accountability.

Electricity Tampering Detection & Smart Energy Meter — *IoT / Embedded*

- Built an IoT-based real-time energy monitoring platform with electricity theft detection using sensors and embedded controllers.

Additional Projects

- Face Detection System (AI/Computer Vision), HVAC Controller (STM32 Bluepill), Exam Hall Allocation System (Embedded), Offline AI Heart Monitor (Digital Twin), Offline AI Coding Assistant, EMI/EMC Hardware Monitoring (STM32), VLSI RTL Design (SystemVerilog + Vivado).

EDUCATION

Bachelor of Engineering – Electronics & Communication Engineering

Velammal Institute of Technology, Chennai | Expected Graduation: 2027 | CGPA: 8.35 (Up to 6th Semester)

Higher Secondary (CBSE)

Cauvery International School, Tiruchirapalli | 80.60%

AWARDS & ACHIEVEMENTS

- ₹12,000+ cumulative prize recognition across multiple technical competitions and symposiums.
- Team Winner — Kalasalingam University (13-member team): secured ₹33,500 top overall cash prize.
- First Prize — VIT 24-Hour Hackathon: resulted in direct internship offer from One Yes Info Tech.
- Three First Prizes across multiple events at AMS College of Engineering; First Prize at Inter-College Project Expo.
- Special Mention — TECHSTROM (LabVIEW myRIO Medical Project); Special Mention — Sir M. Visvesvaraya Institute of Technology Hackathon.
- Finalist — YUKTI Innovation Challenge 2025; Advanced to Round 2 — TN Skills Competition.
- Presented project at IIT Research Park; industrial visits to IIT Communication Laboratory and RETECH.

LANGUAGES

English (Professional) | Tamil (Native)